Introduction to Probability and Statistics

TRAINING QUIZ # 1

Q 1 – Let X be a continuous random variable defined on $\mathbb{R}$ with a probability density function $f_X(x \mid \theta_1, \theta_2)$ where θ_1 and θ_2 are unknown parameters such that $\theta_2 > \theta_1$.

1. Give the expression of the q^{th} theoritical moment :

   $E(X^q) =$

2. Let $X_1, \ldots, X_n$ be n r.v. from the distribution above.
   Give the expression of the q^{th} empirical moment :

   $m_q =$

3. Suppose that $E(X) = \sqrt{\theta_1}$ and $Var(X) = \theta_2 - \theta_1$

   (a) Calculate $E(X^2) =$

(b) Suggest a method to estimate θ_1 and θ_2 and give the estimator of each parameter.

Q 2 – Let $X_1,\ldots,X_n$ be n i.i.d. r.v's with a normal distribution (μ,σ^2).

1. What is the distribution of $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$?

2. An estimator $g_n(\underline{X})$ of a parameter θ is said to be a consistent estimator of θ if it converges in probability to θ that is to say iff $\forall \varepsilon > 0,\ \lim_{n\to+\infty} P\Big(\big|g_n(\underline{X})-\theta\big| > \varepsilon\Big) = 0$.
Using the Chebyshev's Inequality : $\forall \varepsilon > 0,\ P\Big(|Y - EY| \geqslant \varepsilon\Big) \leqslant \frac{Var(Y)}{\varepsilon^2}$, prove that $\overline{X}_n$ is a consistent estimator of μ.

Q 3 – Let $X_1, \ldots, X_n$ be n Bernoulli i.i.d. r.v's with parameter p ($0 < p < 1$).

$$P(X_i = x_i \mid p) = p^{x_i}(1-p)^{(1-x_i)},\ x_i \in \left\{0,1\right\},\ i = 1,\ldots,n.$$

We want to find the m.l.e. of p.

1. Write the likelihood

   $L(p) =$

2. Calculate the log-likelihood

   $\log L(p) =$

3. Give and solve the likelihood equation to express $\hat{p}$ the m.l.e. :

4. Is this estimator unbiased ?